

MASTER PROMPT — Excel VBA Expenses Dashboard v2 (FY 2025)

With Live Progress Tracking, Status Bar, DoEvents Animation & Headings-ON Layout

INSTRUCTION TO AI

You are an expert Excel VBA developer. Generate a **single, complete, self-contained VBA module** that builds a fully automated, professional Expenses Dashboard in Excel.

Non-negotiable rules:

- Output the **entire VBA code in one code block**. No partial code. No placeholders. No "add your logic here" stubs.
 - Reproduce **every Sub, Function, constant, and helper** exactly as specified. Do not merge, rename, omit, or reorder any procedure.
 - Do **not** change any RGB colour value, formula string, range address, chart type enum, slicer style, zoom level, row height, column width, or naming convention.
 - The code must run without errors in Excel for Windows (Office 2016 or later) when the user runs BuildExpensesDashboard.
 - Application.ScreenUpdating must remain **True** throughout so the user sees live progress.
 - The status bar (Application.StatusBar) must display step-by-step progress messages exactly as specified.
-

WORKBOOK PREREQUISITES

Tell the user this before showing the code:

The workbook must have a sheet named **Data** with these exact column headers in Row 1 (order does not matter; all must be present):

Header

Expense ID

Date

Department

Category

Header

Amount (INR)

Payment Method

Approved

Budget Allocated

Budget Remaining

A Month column is **auto-added** by the macro if missing.

MODULE-LEVEL CONSTANTS AND VARIABLES

Option Explicit

Private Const DATA_SHEET As String = "Data"

Private Const PIVOT_SHEET As String = "Pivot"

Private Const DASH_SHEET As String = "Dashboard"

Private Const TABLE_NAME As String = "tblExpenses"

Private Const MONEY_K_FORMAT As String = "#,##0, ""K"""

Private BuildStep As String

ARCHITECTURE — FULL CALL HIERARCHY

BuildExpensesDashboard() ← PUBLIC entry point

 ShowProgress (called throughout)

 PrepareDataSheet

 ShowProgress (×9 sub-steps)

 BuildPivotSheet

 ShowProgress (×10 sub-steps)

 CreatePivot_ExpenseByDepartment

 CreatePivot_ExpenseByCategory

CreatePivot_MonthlyTrend

CreatePivot_ApprovedStatus

CreatePivot_KPI

CreatePivot_PaymentMethod

CreatePivot_BudgetVsActual

BuildChartHelpers

BuildTwoColumnHelper (×4)

BuildBudgetHelper (×1)

BuildDashboardSheet

ShowProgress (×10 sub-steps)

ApplyDashboardCanvas

CreateHeader

CreateKPIHelperFormulas

CreateKPICards

ShowProgress (×6 sub-steps)

AddKPICard (×6)

CreateDashboardCharts

ShowProgress (×5 sub-steps)

AddNormalChart (×4)

AddNormalChartFixedCm (×1)

FormatChart (called by both)

CreateMonthSlicer

CreateDepartmentSlicer

FinalDashboardFormatting

Shared helpers (always include all):

AddPivotToSlicer

PivotDataRowsCount

FormatPivotValues

SafeNumberFormat

DeleteSheetIfExists

LastUsedRow

LastUsedCol

HeaderColumn

RequireHeader

ShowProgress

Wait

COMPLETE VBA CODE — REPRODUCE EXACTLY

SECTION 1 — Progress Helpers

```
Private Sub ShowProgress(ByVal msg As String, _  
    Optional ByVal flashSheet As String = "", _  
    Optional ByVal pauseMs As Long = 350)  
  
    Application.StatusBar = " ⚙ " & msg  
  
    DoEvents  
  
    If flashSheet <> "" Then  
        On Error Resume Next  
        ThisWorkbook.Worksheets(flashSheet).Activate  
        DoEvents  
        On Error GoTo 0  
    End If  
  
    If pauseMs > 0 Then Wait pauseMs  
End Sub
```

```
Private Sub Wait(ByVal ms As Long)  
  
    Dim t As Single  
    t = Timer  
  
    Do While (Timer - t) * 1000 < ms
```

```
DoEvents
Loop
End Sub
```

SECTION 2 — Public Entry Point

Critical behaviour flags:

- Application.ScreenUpdating = True — must stay True so animation is visible
- Application.DisplayAlerts = False
- Application.EnableEvents = False
- Application.Calculation = xlCalculationAutomatic
- On success: clears status bar, shows MsgBox "Dashboard Created Successfully" with vbInformation
- On failure: clears status bar, shows error message with vbCritical

```
Public Sub BuildExpensesDashboard()
```

```
On Error GoTo CleanFail
```

```
BuildStep = "Starting"
```

```
Application.ScreenUpdating = True
```

```
Application.DisplayAlerts = False
```

```
Application.EnableEvents = False
```

```
Application.Calculation = xlCalculationAutomatic
```

```
ShowProgress "Opening Data sheet...", DATA_SHEET, 300
```

```
BuildStep = "Preparing Data"
```

```
ShowProgress "Step 1 of 3 — Formatting Data sheet...", DATA_SHEET, 200
```

```
PrepareDataSheet
```

```
BuildStep = "Building Pivots"
```

ShowProgress "Step 2 of 3 — Building PivotTables...", "", 200

BuildPivotSheet

BuildStep = "Building Dashboard"

ShowProgress "Step 3 of 3 — Building Dashboard...", "", 200

BuildDashboardSheet

BuildStep = "Final Refresh"

ShowProgress "Refreshing all data...", DASH_SHEET, 400

ThisWorkbook.RefreshAll

Application.CalculateFull

Application.Calculation = xlCalculationAutomatic

CleanExit:

Application.StatusBar = False

Application.EnableEvents = True

Application.DisplayAlerts = True

Application.ScreenUpdating = True

MsgBox "Dashboard Created Successfully", vbInformation, "Done"

Exit Sub

CleanFail:

Application.StatusBar = False

Application.Calculation = xlCalculationAutomatic

Application.EnableEvents = True

Application.DisplayAlerts = True

Application.ScreenUpdating = True

MsgBox "Dashboard build failed at step: " & BuildStep & vbCrLf & _

"Error " & Err.Number & ": " & Err.Description, vbCritical

End Sub

SECTION 3 — PrepareDataSheet

Sub activates the Data sheet at start. Every sub-step calls ShowProgress then DoEvents.

Private Sub PrepareDataSheet()

Dim ws As Worksheet, lo As ListObject, rng As Range

Dim lastRow As Long, lastCol As Long

Set ws = ThisWorkbook.Worksheets(DATA_SHEET)

ws.Activate

ShowProgress "Checking required columns...", , 250

RequireHeader ws, "Expense ID"

RequireHeader ws, "Date"

RequireHeader ws, "Department"

RequireHeader ws, "Category"

RequireHeader ws, "Amount (INR)"

RequireHeader ws, "Payment Method"

RequireHeader ws, "Approved"

RequireHeader ws, "Budget Allocated"

RequireHeader ws, "Budget Remaining"

ShowProgress "Inserting Month column...", , 300

If HeaderColumn(ws, "Month") = 0 Then

ws.Cells(1, LastUsedCol(ws) + 1).Value = "Month"

End If

ShowProgress "Creating Excel Table (tblExpenses)...", , 350

```
lastRow = LastUsedRow(ws)
```

```
lastCol = LastUsedCol(ws)
```

```
Set rng = ws.Range(ws.Cells(1, 1), ws.Cells(lastRow, lastCol))
```

```
Dim lo2 As ListObject
```

```
On Error Resume Next
```

```
Set lo2 = ws.ListObjects(TABLE_NAME)
```

```
On Error GoTo 0
```

```
If lo2 Is Nothing Then
```

```
    Set lo = ws.ListObjects.Add(xlSrcRange, rng, , xlYes)
```

```
    lo.Name = TABLE_NAME
```

```
Else
```

```
    Set lo = lo2
```

```
    lo.Resize rng
```

```
End If
```

```
lo.TableStyle = "TableStyleMedium4"
```

```
lo.ShowAutoFilterDropDown = True
```

```
DoEvents
```

```
ShowProgress "Writing Month formula in every row...", , 300
```

```
lo.ListColumns("Month").DataBodyRange.Formula = "=TEXT([@Date],""mmm"")"
```

```
DoEvents
```

```
ShowProgress "Formatting Date column...", , 250
```

```
SafeNumberFormat lo.ListColumns("Date").DataBodyRange, "dd-mmm-yyyy"
```

```
DoEvents
```



```

ShowProgress "Formatting Amount (INR) column...", , 250

SafeNumberFormat lo.ListColumns("Amount (INR)").DataBodyRange,
MONEY_K_FORMAT

DoEvents

ShowProgress "Formatting Budget Allocated column...", , 250

SafeNumberFormat lo.ListColumns("Budget Allocated").DataBodyRange,
MONEY_K_FORMAT

DoEvents

ShowProgress "Formatting Budget Remaining column...", , 250

SafeNumberFormat lo.ListColumns("Budget Remaining").DataBodyRange,
MONEY_K_FORMAT

DoEvents

ShowProgress "Styling header row...", , 300

With ws.Rows(1)
    .Font.Bold = True
    .Interior.Color = RGB(15, 76, 31)
    .Font.Color = RGB(255, 255, 255)
End With

DoEvents

ShowProgress "Auto-fitting column widths...", , 250

ws.Columns.AutoFit

DoEvents

End Sub

```

SECTION 4 — BuildPivotSheet

Order of operations:

1. Delete DASH_SHEET first, then PIVOT_SHEET
2. Add new Pivot sheet after Data sheet
3. Activate it immediately
4. Create PivotCache from tblExpenses using xlR1C1 external address
5. Build all 7 PivotTables in order with ShowProgress before each
6. Build chart helper data
7. AutoFit columns

```
Private Sub BuildPivotSheet()
```

```
    Dim wsP As Worksheet, lo As ListObject, pc As PivotCache
```

```
    ShowProgress "Removing old sheets...", , 300
```

```
    DeleteSheetIfExists DASH_SHEET
```

```
    DeleteSheetIfExists PIVOT_SHEET
```

```
    ShowProgress "Creating Pivot sheet...", , 300
```

```
    Set wsP =
```

```
    ThisWorkbook.Worksheets.Add(After:=ThisWorkbook.Worksheets(DATA_SHEET))
```

```
    wsP.Name = PIVOT_SHEET
```

```
    wsP.Activate
```

```
    DoEvents
```

```
    Set lo = ThisWorkbook.Worksheets(DATA_SHEET).ListObjects(TABLE_NAME)
```

```
    Set pc = ThisWorkbook.PivotCaches.Create( _
```

```
        SourceType:=xlDatabase, _
```

```
        SourceData:=lo.Range.Address(True, True, xlR1C1, True))
```

```
    ShowProgress "Building PivotTable 1/7 — Expense by Department...",  
    PIVOT_SHEET, 400
```

```
    CreatePivot_ExpenseByDepartment pc, wsP.Range("A3")
```

DoEvents

ShowProgress "Building PivotTable 2/7 — Expense by Category...",
PIVOT_SHEET, 400

CreatePivot_ExpenseByCategory pc, wsP.Range("D3")

DoEvents

ShowProgress "Building PivotTable 3/7 — Monthly Trend...", PIVOT_SHEET, 400

CreatePivot_MonthlyTrend pc, wsP.Range("G3")

DoEvents

ShowProgress "Building PivotTable 4/7 — Approved Status...", PIVOT_SHEET,
400

CreatePivot_ApprovedStatus pc, wsP.Range("J3")

DoEvents

ShowProgress "Building PivotTable 5/7 — KPI Totals...", PIVOT_SHEET, 400

CreatePivot_KPI pc, wsP.Range("M3")

DoEvents

ShowProgress "Building PivotTable 6/7 — Payment Method...", PIVOT_SHEET,
400

CreatePivot_PaymentMethod pc, wsP.Range("A25")

DoEvents

ShowProgress "Building PivotTable 7/7 — Budget vs Actual...", PIVOT_SHEET,
400

CreatePivot_BudgetVsActual pc, wsP.Range("D25")

DoEvents

ShowProgress "Generating chart-helper data ranges...", PIVOT_SHEET, 350

BuildChartHelpers wsP

DoEvents

wsP.Columns.AutoFit

End Sub

SECTION 5 — Seven PivotTable Builders

Private Sub CreatePivot_ExpenseByDepartment(ByVal pc As PivotCache, ByVal dest As Range)

Dim pt As PivotTable

Set pt = pc.CreatePivotTable(TableDestination:=dest,
TableName:="PT_ExpenseByDepartment")

With pt

.PivotFields("Department").Orientation = xlRowField

.AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum

.RowAxisLayout xlTabularRow

.RefreshTable

End With

FormatPivotValues pt, MONEY_K_FORMAT

End Sub

Private Sub CreatePivot_ExpenseByCategory(ByVal pc As PivotCache, ByVal dest As Range)

Dim pt As PivotTable

Set pt = pc.CreatePivotTable(TableDestination:=dest,
TableName:="PT_ExpenseByCategory")

With pt

.PivotFields("Category").Orientation = xlRowField

.AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum

```

        .RowAxisLayout xlTabularRow

        .RefreshTable

    End With

    FormatPivotValues pt, MONEY_K_FORMAT

End Sub


Private Sub CreatePivot_MonthlyTrend(ByVal pc As PivotCache, ByVal dest As Range)

    Dim pt As PivotTable

    Set pt = pc.CreatePivotTable(TableDestination:=dest,
    TableName:="PT_MonthlyTrend")

    With pt

        .PivotFields("Month").Orientation = xlRowField

        .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum

        .PivotFields("Month").AutoSort xlAscending, "Month"

        .RowAxisLayout xlTabularRow

        .RefreshTable

    End With

    FormatPivotValues pt, MONEY_K_FORMAT

End Sub


Private Sub CreatePivot_ApprovedStatus(ByVal pc As PivotCache, ByVal dest As Range)

    Dim pt As PivotTable

    Set pt = pc.CreatePivotTable(TableDestination:=dest,
    TableName:="PT_ApprovedStatus")

    With pt

        .PivotFields("Approved").Orientation = xlRowField

        .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum

        .AddDataField .PivotFields("Expense ID", "Count of Expense ID", xlCount

```

```

        .RowAxisLayout xlTabularRow
        .RefreshTable
    End With
    On Error Resume Next
    pt.DataFields(1).DataRange.NumberFormat = MONEY_K_FORMAT
    pt.DataFields(2).DataRange.NumberFormat = "#,##0"
    On Error GoTo 0
End Sub

Private Sub CreatePivot_KPI(ByVal pc As PivotCache, ByVal dest As Range)
    Dim pt As PivotTable
    Set pt = pc.CreatePivotTable(TableDestination:=dest, TableName:="PT_KPI")
    With pt
        .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)",
xlSum
        .AddDataField .PivotFields("Budget Allocated"), "Sum of Budget Allocated",
xlSum
        .AddDataField .PivotFields("Budget Remaining"), "Sum of Budget Remaining",
xlSum
        .AddDataField .PivotFields("Expense ID"), "Count of Expense ID",
xlCount
        .RefreshTable
    End With
    FormatPivotValues pt, MONEY_K_FORMAT
End Sub

```

```

Private Sub CreatePivot_PaymentMethod(ByVal pc As PivotCache, ByVal dest As
Range)
    Dim pt As PivotTable
    Set pt = pc.CreatePivotTable(TableDestination:=dest,
TableName:="PT_PaymentMethod")

```

```

With pt
    .PivotFields("Payment Method").Orientation = xlRowField
    .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum
    .RowAxisLayout xlTabularRow
    .RefreshTable
End With

FormatPivotValues pt, MONEY_K_FORMAT

End Sub

Private Sub CreatePivot_BudgetVsActual(ByVal pc As PivotCache, ByVal dest As
Range)
    Dim pt As PivotTable

    Set pt = pc.CreatePivotTable(TableDestination:=dest,
    TableName:="PT_BudgetVsActual")

    With pt
        .PivotFields("Department").Orientation = xlRowField
        .AddDataField .PivotFields("Amount (INR)", "Actual Spent", xlSum
        .AddDataField .PivotFields("Budget Allocated"), "Sum of Budget Allocated",
xlSum
        .RowAxisLayout xlTabularRow
        .RefreshTable
    End With

    FormatPivotValues pt, MONEY_K_FORMAT

End Sub

```

SECTION 6 — Chart Helper Data Builders

```

Private Sub BuildChartHelpers(ByVal ws As Worksheet)

    ws.Range("A50:Z95").ClearContents

```

```
BuildTwoColumnHelper ws, "A50", ws.PivotTables("PT_ExpenseByDepartment"),  
"Department", "Amount"
```

```
BuildTwoColumnHelper ws, "D50", ws.PivotTables("PT_ExpenseByCategory"),  
"Category", "Amount"
```

```
BuildTwoColumnHelper ws, "G50", ws.PivotTables("PT_MonthlyTrend"),  
"Month", "Amount"
```

```
BuildTwoColumnHelper ws, "J50", ws.PivotTables("PT_PaymentMethod"),  
"Payment Method", "Amount"
```

```
BuildBudgetHelper ws, "M50", ws.PivotTables("PT_BudgetVsActual")
```

```
End Sub
```

```
Private Sub BuildTwoColumnHelper(ByVal ws As Worksheet, ByVal topLeft As  
String, _
```

```
ByVal pt As PivotTable, ByVal h1 As String, ByVal h2 As String)
```

```
Dim r As Range, i As Long, n As Long
```

```
Set r = ws.Range(topLeft)
```

```
r.Value = h1
```

```
r.Offset(0, 1).Value = h2
```

```
n = PivotDataRowsCount(pt)
```

```
For i = 1 To n
```

```
    r.Offset(i, 0).Formula = "=" & pt.RowRange.Cells(i + 1, 1).Address(True, True,  
xlA1, True)
```

```
    r.Offset(i, 1).Formula = "=" & pt.DataBodyRange.Cells(i, 1).Address(True, True,  
xlA1, True)
```

```
Next i
```

```
SafeNumberFormat r.Offset(1, 1).Resize(n, 1), MONEY_K_FORMAT
```

```
End Sub
```

```
Private Sub BuildBudgetHelper(ByVal ws As Worksheet, ByVal topLeft As String,  
ByVal pt As PivotTable)
```

```
Dim r As Range, i As Long, n As Long
```

```
Set r = ws.Range(topLeft)
```



```

r.Value = "Department"
r.Offset(0, 1).Value = "Actual Spent"
r.Offset(0, 2).Value = "Budget Allocated"
n = PivotDataRowsCount(pt)
For i = 1 To n
    r.Offset(i, 0).Formula = "=" & pt.RowRange.Cells(i + 1, 1).Address(True, True, xlA1, True)
    r.Offset(i, 1).Formula = "=" & pt.DataBodyRange.Cells(i, 1).Address(True, True, xlA1, True)
    r.Offset(i, 2).Formula = "=" & pt.DataBodyRange.Cells(i, 2).Address(True, True, xlA1, True)
Next i
SafeNumberFormat r.Offset(1, 1).Resize(n, 2), MONEY_K_FORMAT
End Sub

```

SECTION 7 — BuildDashboardSheet Orchestrator

Critical sequence — must not be reordered:

1. Delete old DASH_SHEET, add new one Before sheet index 1, name it, Activate it
2. Set ActiveWindow.Zoom = 70 and ActiveWindow.DisplayHeadings = True **immediately after Activate**, before any drawing
3. Call all sub-procedures with ShowProgress + DoEvents between each
4. At the very end, re-Activate Dashboard, re-set Zoom = 70, re-set DisplayHeadings = True as the final authoritative state

```

Private Sub BuildDashboardSheet()
    Dim wsD As Worksheet

    ShowProgress "Creating Dashboard sheet...", , 300
    DeleteSheetIfExists DASH_SHEET
    Set wsD = ThisWorkbook.Worksheets.Add(Before:=ThisWorkbook.Worksheets(1))
    wsD.Name = DASH_SHEET

```

wsD.Activate

' — Set zoom and turn headings ON immediately — before anything is drawn —

ActiveWindow.Zoom = 70

ActiveWindow.DisplayHeadings = True

DoEvents

ShowProgress "Applying canvas & grid settings...",
DASH_SHEET, 300

ApplyDashboardCanvas wsD

DoEvents

ShowProgress "Drawing header banner...",
400 DASH_SHEET,

CreateHeader wsD

DoEvents

ShowProgress "Writing KPI helper formulas...",
300 DASH_SHEET,

CreateKPIHelperFormulas wsD

DoEvents

ShowProgress "Building KPI cards (6 cards)...",
500 DASH_SHEET,

CreateKPICards wsD

DoEvents

ShowProgress "Inserting charts — Department, Category, Trend, Budget,
Payment...", DASH_SHEET, 600

CreateDashboardCharts wsD

DoEvents

```
ShowProgress "Adding Month slicer...", DASH_SHEET,
400
```

```
CreateMonthSlicer wsD
```

```
DoEvents
```

```
ShowProgress "Adding Department slicer...", DASH_SHEET,
400
```

```
CreateDepartmentSlicer wsD
```

```
DoEvents
```

```
ShowProgress "Applying final formatting & print settings...",
DASH_SHEET, 350
```

```
FinalDashboardFormatting wsD
```

```
DoEvents
```

```
' — Re-activate Dashboard and permanently lock in headings ON at 70%
```

```
wsD.Activate
```

```
wsD.Range("A1").Select
```

```
ActiveWindow.Zoom = 70
```

```
ActiveWindow.DisplayHeadings = True ' final authoritative set — headings stay
ON permanently
```

```
End Sub
```

SECTION 8 — ApplyDashboardCanvas

Note: DisplayHeadings is intentionally **NOT** touched here. It is controlled only in BuildDashboardSheet.

```
Private Sub ApplyDashboardCanvas(ByVal ws As Worksheet)
```

```
Dim i As Long
```

```
ws.Activate
```

```

ActiveWindow.DisplayGridlines = False

' DisplayHeadings is NOT touched here — controlled only in
BuildDashboardSheet

```

```

ws.Cells.Interior.Color = RGB(246, 250, 244)
ws.Cells.Font.Name = "Aptos"
ws.Cells.Font.Size = 10

```

```

For i = 1 To 30
    ws.Columns(i).ColumnWidth = 7.3
Next i

```

```

ws.Rows("1:3").RowHeight = 22
ws.Rows("4:29").RowHeight = 18
End Sub

```

SECTION 9 — CreateHeader

All positions calculated from live range measurements — no hardcoded pixel values for banner/title/icon. Only the date box uses a fixed width (190) and the icon uses fixed size (42×42).

Shape Key measurements

Banner Left/Top/Width/Height from ws.Range("A1:AD3")

Icon oval 42×42, Left = bannerLeft+12, Top = bannerTop+(bannerHeight-42)/2,
Font size 22

Title Left = iconLeft+iconW+8, Width = dateLeft-titleLeft-8, full banner height,
textbox Font size 44

Date Width=190, Left = bannerLeft+bannerWidth-190-10, Top = bannerTop+4,
textbox Height = bannerHeight-8, Font size 18

```
Private Sub CreateHeader(ByVal ws As Worksheet)
```

```
    Dim shp As Shape, icon As Shape, dateBox As Shape, titleBox As Shape
```

```
Dim bannerLeft As Double: bannerLeft = ws.Range("A1").Left
Dim bannerTop As Double: bannerTop = ws.Range("A1").Top
Dim bannerWidth As Double: bannerWidth = ws.Range("A1:AD3").Width
Dim bannerHeight As Double: bannerHeight = ws.Range("A1:AD3").Height
```

```
Dim iconW As Double: iconW = 42
Dim iconH As Double: iconH = 42
Dim iconLeft As Double: iconLeft = bannerLeft + 12
Dim iconTop As Double: iconTop = bannerTop + (bannerHeight - iconH) / 2
```

```
Dim dateW As Double: dateW = 190
Dim dateH As Double: dateH = bannerHeight - 8
Dim dateLeft As Double: dateLeft = bannerLeft + bannerWidth - dateW - 10
Dim dateTop As Double: dateTop = bannerTop + 4
```

```
Dim titleLeft As Double: titleLeft = iconLeft + iconW + 8
Dim titleWidth As Double: titleWidth = dateLeft - titleLeft - 8
Dim titleTop As Double: titleTop = bannerTop
Dim titleHeight As Double: titleHeight = bannerHeight
```

```
' === Green gradient banner ===
```

```
Set shp = ws.Shapes.AddShape(msoShapeRoundedRectangle, bannerLeft,
bannerTop, bannerWidth, bannerHeight)
```

```
With shp
```

```
.Name = "Header_Banner"
```

```
.Fill.TwoColorGradient msoGradientHorizontal, 1
```

```
.Fill.ForeColor.RGB = RGB(6, 66, 25)
```

```
.Fill.BackColor.RGB = RGB(17, 112, 31)
```

```
.Line.Visible = msoFalse
```

```
.TextFrame.Characters.Text = ""
```

```
End With
```

```
' === Dollar-sign oval icon ===
```

```
Set icon = ws.Shapes.AddShape(msoShapeOval, iconLeft, iconTop, iconW,  
iconH)
```

```
With icon
```

```
.Fill.ForeColor.RGB = RGB(12, 88, 30)
```

```
.Line.ForeColor.RGB = RGB(230, 245, 230)
```

```
.TextFrame.Characters.Text = "$"
```

```
.TextFrame.HorizontalAlignment = xlHAlignCenter
```

```
.TextFrame.VerticalAlignment = xlVAlignCenter
```

```
.TextFrame.Characters.Font.Color = RGB(255, 255, 255)
```

```
.TextFrame.Characters.Font.Bold = True
```

```
.TextFrame.Characters.Font.Size = 22
```

```
End With
```

```
' === Title textbox (centre) ===
```

```
Set titleBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _  
titleLeft, titleTop, titleWidth, titleHeight)
```

```
With titleBox
```

```
.Fill.Visible = msoFalse
```

```
.Line.Visible = msoFalse
```

```
.TextFrame2.WordWrap = msoFalse
```

```
.TextFrame2.AutoSize = msoAutoSizeNone
```

```
.TextFrame.Characters.Text = "EXPENSES DASHBOARD | FY 2025"
```

```
.TextFrame.HorizontalAlignment = xlHAlignCenter
```

```
.TextFrame.VerticalAlignment = xlVAlignCenter
```

```

        .TextFrame.Characters.Font.Color = RGB(255, 255, 255)
        .TextFrame.Characters.Font.Bold = True
        .TextFrame.Characters.Font.Size = 44
    End With

' === Date textbox (right) ===
Set dateBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _
    dateLeft, dateTop, dateW, dateH)
With dateBox
    .Fill.Visible = msoFalse
    .Line.Visible = msoFalse
    .TextFrame2.WordWrap = msoTrue
    .TextFrame2.AutoSize = msoAutoSizeNone
    .TextFrame.Characters.Text = "Data as of:" & vbCrLf & Format(Date, "mmm dd,
yyyy")
    .TextFrame.HorizontalAlignment = xlHAlignCenter
    .TextFrame.VerticalAlignment = xlVAlignCenter
    .TextFrame.Characters.Font.Color = RGB(255, 255, 255)
    .TextFrame.Characters.Font.Bold = True
    .TextFrame.Characters.Font.Size = 18
End With
End Sub

```

SECTION 10 — KPI Helper Formulas

All formulas use GETPIVOTDATA referencing Pivot!\$M\$3 (PT_KPI) and Pivot!\$J\$3 (PT_ApprovedStatus). Columns AF:AG are hidden at the end.

```

Private Sub CreateKPIHelperFormulas(ByVal ws As Worksheet)
    ws.Range("AF1:AG20").ClearContents

```

```

ws.Range("AF2").Formula = "=IFERROR(GETPIVOTDATA(""Sum of Amount (INR)"" ,Pivot!$M$3),0)"

ws.Range("AF3").Formula = "=IFERROR(GETPIVOTDATA(""Sum of Budget Allocated"" ,Pivot!$M$3),0)"

ws.Range("AF4").Formula = "=IFERROR(GETPIVOTDATA(""Sum of Budget Remaining"" ,Pivot!$M$3),0)"

ws.Range("AF5").Formula = "=IFERROR(GETPIVOTDATA(""Count of Expense ID"" ,Pivot!$M$3),0)"

ws.Range("AF6").Formula = "=IFERROR(GETPIVOTDATA(""Count of Expense ID"" ,Pivot!$J$3, ""Approved"" , ""Yes"" )/AF5,0)"

ws.Range("AF7").Formula = "=IFERROR(AF2/AF5,0)"

SafeNumberFormat ws.Range("AF2:AF4,AF7"), MONEY_K_FORMAT

SafeNumberFormat ws.Range("AF5"),      "#,##0"

SafeNumberFormat ws.Range("AF6"),      "0.0%"

End Sub

```

SECTION 11 — KPI Cards

Six cards. Each gets its own ShowProgress call + DoEvents. Cards use formula-linked textboxes for live values (updated by slicers).

Address	Title	Formula	Icon	Sub-label	Accent RGB
A5:E8	Total Expenses	=Dashboard!\$AF\$2 \$		Total Spend	RGB(21,125,24)
F5:J8	Budget Allocated	=Dashboard!\$AF\$3 %		100% of Plan	RGB(67,152,32)
K5:O8	Budget Remaining	=Dashboard!\$AF\$4 \$		Remaining Budget	RGB(93,171,57)
P5:T8	Transactions	=Dashboard!\$AF\$5 #		Total Transactions	RGB(17,121,58)
U5:Y8	Approval Rate	=Dashboard!\$AF\$6 O		Approved Items	RGB(37,137,48)

Address	Title	Formula	Icon	Sub-label	Accent	RGB
Z5:AD8	Avg Expense	=Dashboard!\$AF\$7 ^		Per Transaction		RGB(22,101,52)

Private Sub CreateKPICards(ByVal ws As Worksheet)

 ShowProgress " KPI card 1/6 — Total Expenses...", DASH_SHEET, 300

 AddKPICard ws, "A5:E8", "Total Expenses", "=Dashboard!\$AF\$2", "\$", "Total Spend", RGB(21, 125, 24)

 DoEvents

 ShowProgress " KPI card 2/6 — Budget Allocated...", DASH_SHEET, 300

 AddKPICard ws, "F5:J8", "Budget Allocated", "=Dashboard!\$AF\$3", "%", "100% of Plan", RGB(67, 152, 32)

 DoEvents

 ShowProgress " KPI card 3/6 — Budget Remaining...", DASH_SHEET, 300

 AddKPICard ws, "K5:O8", "Budget Remaining", "=Dashboard!\$AF\$4", "\$", "Remaining Budget", RGB(93, 171, 57)

 DoEvents

 ShowProgress " KPI card 4/6 — Transactions...", DASH_SHEET, 300

 AddKPICard ws, "P5:T8", "Transactions", "=Dashboard!\$AF\$5", "#", "Total Transactions", RGB(17, 121, 58)

 DoEvents

 ShowProgress " KPI card 5/6 — Approval Rate...", DASH_SHEET, 300

 AddKPICard ws, "U5:Y8", "Approval Rate", "=Dashboard!\$AF\$6", "O", "Approved Items", RGB(37, 137, 48)

 DoEvents

 ShowProgress " KPI card 6/6 — Avg Expense...", DASH_SHEET, 300

 AddKPICard ws, "Z5:AD8", "Avg Expense", "=Dashboard!\$AF\$7", "^", "Per Transaction", RGB(22, 101, 52)

 DoEvents

End Sub

```
Private Sub AddKPICard(ByVal ws As Worksheet, ByVal addr As String, ByVal title  
As String, _
```

```
    ByVal valueFormula As String, ByVal iconText As String, _
```

```
    ByVal subTitle As String, ByVal accent As Long)
```

```
Dim box As Range, card As Shape, icon As Shape
```

```
Dim titleBox As Shape, valueBox As Shape, subBox As Shape
```

```
Set box = ws.Range(addr)
```

```
' White card with soft green border + shadow
```

```
Set card = ws.Shapes.AddShape(msoShapeRoundedRectangle, box.Left,  
box.Top, box.Width, box.Height)
```

```
With card
```

```
    .Fill.ForeColor.RGB = RGB(255, 255, 255)
```

```
    .Line.ForeColor.RGB = RGB(208, 226, 203)
```

```
    .Line.Weight = 0.75
```

```
On Error Resume Next
```

```
    .Shadow.Visible = msoTrue
```

```
    .Shadow.ForeColor.RGB = RGB(200, 210, 200)
```

```
    .Shadow.Transparency = 0.7
```

```
    .Shadow.OffsetX = 1
```

```
    .Shadow.OffsetY = 1
```

```
On Error GoTo 0
```

```
End With
```

```
' Gradient oval icon (38×38, offset 10/16 inside card)
```

```
Set icon = ws.Shapes.AddShape(msoShapeOval, box.Left + 10, box.Top + 16, 38,  
38)
```

```
With icon
```

```
    .Fill.TwoColorGradient msoGradientDiagonalUp, 1
```

```
.Fill.ForeColor.RGB = accent
.Fill.BackColor.RGB = RGB(8, 80, 25)
.Line.Visible = msoFalse
.TextFrame.Characters.Text = iconText
.TextFrame.HorizontalAlignment = xlHAlignCenter
.TextFrame.VerticalAlignment = xlVAlignCenter
.TextFrame.Characters.Font.Color = RGB(255, 255, 255)
.TextFrame.Characters.Font.Bold = True
.TextFrame.Characters.Font.Size = 13
```

End With

' Title label (offset 54/8, height 22, size 18 bold dark green)

```
Set titleBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _
    box.Left + 54, box.Top + 8, box.Width - 58, 22)
```

With titleBox

```
.Fill.Visible = msoFalse
.Line.Visible = msoFalse
.TextFrame.Characters.Text = title
.TextFrame.HorizontalAlignment = xlHAlignCenter
.TextFrame.VerticalAlignment = xlVAlignCenter
.TextFrame.Characters.Font.Bold = True
.TextFrame.Characters.Font.Size = 18
.TextFrame.Characters.Font.Color = RGB(45, 65, 45)
```

End With

' Value textbox — formula-linked, offset 54/29, height 34, size 28 bold deep green

```
Set valueBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _
    box.Left + 54, box.Top + 29, box.Width - 58, 34)
```

With valueBox

```

.Fill.Visible = msoFalse
.Line.Visible = msoFalse
.TextFrame.HorizontalAlignment = xlHAlignCenter
.TextFrame.VerticalAlignment = xlVAlignCenter
.DrawingObject.Formula = valueFormula
.TextFrame.Characters.Font.Bold = True
.TextFrame.Characters.Font.Size = 28
.TextFrame.Characters.Font.Color = RGB(5, 74, 25)

```

End With

' Subtitle label (offset 54/58, height 13, size 8, grey)

```

Set subBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _
    box.Left + 54, box.Top + 58, box.Width - 58, 13)

```

With subBox

```

.Fill.Visible = msoFalse
.Line.Visible = msoFalse
.TextFrame.Characters.Text = subTitle
.TextFrame.HorizontalAlignment = xlHAlignCenter
.TextFrame.Characters.Font.Size = 8
.TextFrame.Characters.Font.Color = RGB(80, 95, 80)

```

End With

End Sub

SECTION 12 — Dashboard Charts

Five charts. Each gets ShowProgress + DoEvents.

#	Title	Type	Source helper	Target range	chartKind	Size
1	Expense by Department	xlColumnClustered	Pivot!A50	A10:L18	dept	Normal

#	Title	Type	Source helper	Target range	chartKind	Size
2	Expense by Category	xlDoughnut	Pivot!D50	M10:V18	category	Normal
3	Monthly Trend	xlLineMarkers	Pivot!G50	A19:L28	trend	Normal
4	Budget vs Actual	xlColumnClustered	Pivot!M50	M19:V28	budget	Normal
5	Payment Method	xlBarClustered	Pivot!J50	W19:AD28	payment	7.18 cm height, then ×0.8986 shrink

Private Sub CreateDashboardCharts(ByVal ws As Worksheet)

Dim p As Worksheet

Set p = ThisWorkbook.Worksheets(PIVOT_SHEET)

ShowProgress " Chart 1/5 — Expense by Department...", DASH_SHEET, 450

AddNormalChart ws, xlColumnClustered, "Expense by Department",
p.Range("A50").CurrentRegion, "A10:L18", "dept"

DoEvents

ShowProgress " Chart 2/5 — Expense by Category...", DASH_SHEET, 450

AddNormalChart ws, xlDoughnut, "Expense by Category",
p.Range("D50").CurrentRegion, "M10:V18", "category"

DoEvents

ShowProgress " Chart 3/5 — Monthly Trend...", DASH_SHEET, 450

AddNormalChart ws, xlLineMarkers, "Monthly Trend",
p.Range("G50").CurrentRegion, "A19:L28", "trend"

DoEvents

ShowProgress " Chart 4/5 — Budget vs Actual...", DASH_SHEET, 450

```
AddNormalChart ws, xlColumnClustered, "Budget vs Actual",  
p.Range("M50").CurrentRegion, "M19:V28", "budget"
```

```
DoEvents
```

```
ShowProgress " Chart 5/5 — Payment Method...", DASH_SHEET, 450
```

```
AddNormalChartFixedCm ws, xlBarClustered, "Payment Method",  
p.Range("J50").CurrentRegion, "W19:AD28", "payment", 7.18
```

```
DoEvents
```

```
End Sub
```

```
Private Sub AddNormalChart(ByVal ws As Worksheet, ByVal chartType As  
xlChartType, ByVal chartTitle As String, _  
ByVal sourceRange As Range, ByVal targetAddress As String,  
ByVal chartKind As String)
```

```
Dim box As Range, co As ChartObject
```

```
Set box = ws.Range(targetAddress)
```

```
Set co = ws.ChartObjects.Add(box.Left, box.Top, box.Width, box.Height)
```

```
FormatChart co, chartType, chartTitle, sourceRange, chartKind
```

```
End Sub
```

```
Private Sub AddNormalChartFixedCm(ByVal ws As Worksheet, ByVal chartType As  
xlChartType, ByVal chartTitle As String, _
```

```
ByVal sourceRange As Range, ByVal targetAddress As String,
```

```
—
```

```
ByVal chartKind As String, ByVal heightCm As Double)
```

```
Dim box As Range, co As ChartObject
```

```
Set box = ws.Range(targetAddress)
```

```
Set co = ws.ChartObjects.Add(box.Left, box.Top, box.Width,  
Application.CentimetersToPoints(heightCm))
```

```
FormatChart co, chartType, chartTitle, sourceRange, chartKind
```

```
End Sub
```

SECTION 13 — FormatChart (Master Chart Formatter)

Legend rules:

- "category" → xlLegendPositionRight, Font.Size = 12
- "payment" → HasLegend = False
- All others → xlLegendPositionBottom

Doughnut category legend — use fixed pixel values (not ratio) for Left and Width:

.Legend.Left = 229.614

.Legend.Width = 206.285

(Top and Height still use ratio: ChartArea.Height * 0.13 / * 0.75)

Payment chart post-format height adjustment: After the With co.Chart block closes, apply:

If chartKind = "payment" Then

 co.Height = co.Height * 0.8986011428

End If

Private Sub FormatChart(ByVal co As ChartObject, ByVal chartType As XlChartType, ByVal chartTitle As String, _

 ByVal sourceRange As Range, ByVal chartKind As String)

 Dim s As Series, i As Long

 Dim colors As Variant

 colors = Array(RGB(10,93,30), RGB(132,198,55), RGB(93,177,58), RGB(22,138,150), _

 RGB(112,48,160), RGB(97,135,36), RGB(145,160,170), RGB(237,125,49))

With co.Chart

 .ChartType = chartType

 .SetSourceData Source:=sourceRange

 .PlotVisibleOnly = False

.HasTitle = True

.ChartTitle.Text = chartTitle

On Error Resume Next

.ChartTitle.Format.TextFrame2.TextRange.Font.Size = 12

.ChartTitle.Format.TextFrame2.TextRange.Font.Bold = msoTrue

.ChartTitle.Format.TextFrame2.TextRange.Font.Fill.ForeColor.RGB = RGB(5, 74, 25)

.ChartArea.Format.Fill.ForeColor.RGB = RGB(255, 255, 255)

.PlotArea.Format.Fill.ForeColor.RGB = RGB(255, 255, 255)

.ChartArea.Format.Line.ForeColor.RGB = RGB(208, 226, 203)

.ChartArea.Format.Line.Weight = 0.7

.HasLegend = True

If chartKind = "category" Then

.Legend.Position = xlLegendPositionRight

.Legend.Font.Size = 12

Elseif chartKind = "payment" Then

.HasLegend = False

Else

.Legend.Position = xlLegendPositionBottom

End If

If chartKind <> "category" Then

.Axes(xlValue).TickLabels.NumberFormat = MONEY_K_FORMAT

.Axes(xlValue).Format.Line.ForeColor.RGB = RGB(210, 210, 210)

.Axes(xlCategory).Format.Line.ForeColor.RGB = RGB(210, 210, 210)

.Axes(xlValue).MajorGridlines.Format.Line.ForeColor.RGB = RGB(230, 235, 230)

.Axes(xlValue).MajorGridlines.Format.Line.Weight = 0.5

End If

For Each s In .SeriesCollection

s.HasDataLabels = True

s.DataLabels.NumberFormat = MONEY_K_FORMAT

s.DataLabels.Font.Size = 8

s.DataLabels.Font.Bold = True

s.DataLabels.Font.Color = RGB(55, 55, 55)

Next s

Select Case chartKind

Case "dept"

For Each s In .SeriesCollection

s.Format.Fill.TwoColorGradient msoGradientVertical, 1

s.Format.Fill.ForeColor.RGB = RGB(10, 93, 30)

s.Format.Fill.BackColor.RGB = RGB(39, 142, 41)

s.Format.Line.ForeColor.RGB = RGB(6, 66, 25)

Next s

Case "category"

.ChartGroups(1).DoughnutHoleSize = 40

For Each s In .SeriesCollection

For i = 1 To s.Points.Count

s.Points(i).Format.Fill.ForeColor.RGB = colors((i - 1) Mod 8)

s.Points(i).Format.Line.ForeColor.RGB = RGB(255, 255, 255)

Next i

s.HasDataLabels = True

s.DataLabels.ShowValue = True

```
s.DataLabels.ShowPercentage = False
s.DataLabels.ShowCategoryName = False
s.DataLabels.Position = xlLabelPositionBestFit
s.DataLabels.NumberFormat = MONEY_K_FORMAT
s.DataLabels.Font.Size = 8
s.DataLabels.Font.Bold = True
s.DataLabels.Font.Color = RGB(255, 255, 255)
```

Next s

```
.PlotArea.Left = .ChartArea.Width * 0.03
.PlotArea.Top = .ChartArea.Height * 0.12
.PlotArea.Width = .ChartArea.Width * 0.55
.PlotArea.Height = .ChartArea.Height * 0.78
.Legend.Left = 229.614
.Legend.Width = 206.285
.Legend.Top = .ChartArea.Height * 0.13
.Legend.Height = .ChartArea.Height * 0.75
```

Case "trend"

For Each s In .SeriesCollection

```
s.Format.Line.ForeColor.RGB = RGB(8, 88, 28)
s.Format.Line.Weight = 2.25
s.MarkerStyle = xlMarkerStyleCircle
s.MarkerSize = 6
s.MarkerForegroundColor = RGB(8, 88, 28)
s.MarkerBackgroundColor = RGB(39, 142, 41)
s.DataLabels.Font.Color = RGB(5, 74, 25)
```

Next s

Case "budget"

```

        If .SeriesCollection.Count >= 1 Then
        .SeriesCollection(1).Format.Fill.ForeColor.RGB = RGB(10, 93, 30)

        If .SeriesCollection.Count >= 2 Then
        .SeriesCollection(2).Format.Fill.ForeColor.RGB = RGB(166, 210, 96)

```

```

Case "payment"

```

```

    On Error Resume Next

```

```

    .ChartGroups(1).GapWidth = 70

```

```

    On Error GoTo 0

```

```

    For Each s In .SeriesCollection

```

```

        s.Format.Fill.TwoColorGradient msoGradientHorizontal, 1

```

```

        s.Format.Fill.ForeColor.RGB = RGB(10, 93, 30)

```

```

        s.Format.Fill.BackColor.RGB = RGB(39, 142, 41)

```

```

        s.DataLabels.Position = xlLabelPositionOutsideEnd

```

```

    Next s

```

```

End Select

```

```

    On Error GoTo 0

```

```

End With

```

```

If chartKind = "payment" Then

```

```

    co.Height = co.Height * 0.8986011428

```

```

End If

```

```

End Sub

```

SECTION 14 — Month Slicer

- Position: W10 to AD14
- 4 columns, caption = "Month", style = SlicerStyleLight6
- Connected to all 7 PivotTables

Private Sub CreateMonthSlicer(ByVal wsD As Worksheet)

Dim sc As SlicerCache, sl As Slicer, pt As PivotTable

Dim t As Double, l As Double, w As Double, h As Double

On Error Resume Next

ThisWorkbook.SlicerCaches("Slicer_Month").Delete

On Error GoTo 0

Set pt =

ThisWorkbook.Worksheets(PIVOT_SHEET).PivotTables("PT_ExpenseByDepartment")

On Error Resume Next

Set sc = ThisWorkbook.SlicerCaches.Add2(pt, "Month", "Slicer_Month")

If sc Is Nothing Then Set sc = ThisWorkbook.SlicerCaches.Add(pt, "Month", "Slicer_Month")

On Error GoTo 0

If sc Is Nothing Then Exit Sub

AddPivotToSlicer sc, "PT_ExpenseByDepartment"

AddPivotToSlicer sc, "PT_ExpenseByCategory"

AddPivotToSlicer sc, "PT_MonthlyTrend"

AddPivotToSlicer sc, "PT_ApprovedStatus"

AddPivotToSlicer sc, "PT_KPI"

AddPivotToSlicer sc, "PT_PaymentMethod"

AddPivotToSlicer sc, "PT_BudgetVsActual"

t = wsD.Range("W10").Top

l = wsD.Range("W10").Left

```
w = wsD.Range("W10:AD14").Width
h = wsD.Range("W10:AD14").Height
```

```
On Error Resume Next
```

```
Set sl = sc.Slicers.Add(wsD, , "Month", "Month", t, l, w, h)
```

```
If Not sl Is Nothing Then
```

```
    sl.Top = t : sl.Left = l : sl.Width = w : sl.Height = h
```

```
    sl.NumberOfColumns = 4
```

```
    sl.Caption = "Month"
```

```
    sl.Style = "SlicerStyleLight6"
```

```
    sl.Shape.Fill.ForeColor.RGB = RGB(247, 252, 245)
```

```
    sl.Shape.Line.ForeColor.RGB = RGB(184, 215, 176)
```

```
End If
```

```
On Error GoTo 0
```

```
End Sub
```

SECTION 15 — Department Slicer

- Position: W15 to AD18
- 6 columns, caption = "Department", style = SlicerStyleLight6
- Connected to all 7 PivotTables

```
Private Sub CreateDepartmentSlicer(ByVal wsD As Worksheet)
```

```
    Dim sc As SlicerCache, sl As Slicer, pt As PivotTable
```

```
    Dim t As Double, l As Double, w As Double, h As Double
```

```
On Error Resume Next
```

```
ThisWorkbook.SlicerCaches("Slicer_Department").Delete
```

```
On Error GoTo 0
```

```
Set pt =  
ThisWorkbook.Worksheets(PIVOT_SHEET).PivotTables("PT_ExpenseByDepartment")
```

```
On Error Resume Next
```

```
Set sc = ThisWorkbook.SlicerCaches.Add2(pt, "Department",  
"Slicer_Department")
```

```
If sc Is Nothing Then Set sc = ThisWorkbook.SlicerCaches.Add(pt, "Department",  
"Slicer_Department")
```

```
On Error GoTo 0
```

```
If sc Is Nothing Then Exit Sub
```

```
AddPivotToSlicer sc, "PT_ExpenseByDepartment"
```

```
AddPivotToSlicer sc, "PT_ExpenseByCategory"
```

```
AddPivotToSlicer sc, "PT_MonthlyTrend"
```

```
AddPivotToSlicer sc, "PT_ApprovedStatus"
```

```
AddPivotToSlicer sc, "PT_KPI"
```

```
AddPivotToSlicer sc, "PT_PaymentMethod"
```

```
AddPivotToSlicer sc, "PT_BudgetVsActual"
```

```
t = wsD.Range("W15").Top
```

```
l = wsD.Range("W15").Left
```

```
w = wsD.Range("W15:AD18").Width
```

```
h = wsD.Range("W15:AD18").Height
```

```
On Error Resume Next
```

```
Set sl = sc.Slicers.Add(wsD, , "Department", "Department", t, l, w, h)
```

```
If Not sl Is Nothing Then
```

```
sl.Top = t : sl.Left = l : sl.Width = w : sl.Height = h
```

```

sl.NumberOfColumns = 6
sl.Caption = "Department"
sl.Style = "SlicerStyleLight6"
sl.Shape.Fill.ForeColor.RGB = RGB(247, 252, 245)
sl.Shape.Line.ForeColor.RGB = RGB(184, 215, 176)
End If
On Error GoTo 0
End Sub

```

SECTION 16 — FinalDashboardFormatting

Note: DisplayHeadings is intentionally **NOT set here**. It is set to True at the very end of BuildDashboardSheet so it permanently stays ON.

```

Private Sub FinalDashboardFormatting(ByVal ws As Worksheet)
    ws.Range("A1:AD29").Font.Name = "Aptos"
    ws.Range("A1:AD29").VerticalAlignment = xlCenter
    ws.Columns("AF:AG").Hidden = True

    ActiveWindow.DisplayGridlines = False
    ' DisplayHeadings is intentionally NOT set here — see BuildDashboardSheet end

    ws.PageSetup.Zoom = False
    ws.PageSetup.FitToPagesWide = 1
    ws.PageSetup.FitToPagesTall = 1
End Sub

```

SECTION 17 — Shared Utility Functions

```

Private Sub AddPivotToSlicer(ByVal sc As SlicerCache, ByVal ptName As String)
    On Error Resume Next
    sc.PivotTables.AddPivotTable
    ThisWorkbook.Worksheets(PIVOT_SHEET).PivotTables(ptName)

```

```

    On Error GoTo 0
End Sub

Private Function PivotDataRowsCount(ByVal pt As PivotTable) As Long
    Dim n As Long
    On Error Resume Next
    n = pt.DataBodyRange.Rows.Count
    On Error GoTo 0
    If n <= 1 Then PivotDataRowsCount = 1 Else PivotDataRowsCount = n - 1
End Function

Private Sub FormatPivotValues(ByVal pt As PivotTable, ByVal fmt As String)
    Dim df As PivotField
    On Error Resume Next
    If Not pt.DataBodyRange Is Nothing Then pt.DataBodyRange.NumberFormat =
fmt
    For Each df In pt.DataFields
        If InStr(1, df.Caption, "Count", vbTextCompare) > 0 Then
            df.DataRange.NumberFormat = "#,##0"
        Else
            df.DataRange.NumberFormat = fmt
        End If
    Next df
    pt.TableRange1.Font.Name = "Aptos"
    pt.TableRange1.Font.Size = 10
    On Error GoTo 0
End Sub

Private Sub SafeNumberFormat(ByVal rng As Range, ByVal fmt As String)

```



```
On Error Resume Next
rng.NumberFormat = fmt
On Error GoTo 0
End Sub
```

```
Private Sub DeleteSheetIfExists(ByVal sheetName As String)

    Dim ws As Worksheet

    On Error Resume Next

    Set ws = ThisWorkbook.Worksheets(sheetName)

    On Error GoTo 0

    If Not ws Is Nothing Then ws.Delete

End Sub
```

```
Private Function LastUsedRow(ByVal ws As Worksheet) As Long

    Dim c As Range

    Set c = ws.Cells.Find(What:="", LookIn:=xlFormulas, SearchOrder:=xlByRows,
SearchDirection:=xlPrevious)

    If c Is Nothing Then LastUsedRow = 1 Else LastUsedRow = c.Row

End Function
```

```
Private Function LastUsedCol(ByVal ws As Worksheet) As Long

    Dim c As Range

    Set c = ws.Cells.Find(What:="", LookIn:=xlFormulas,
SearchOrder:=xlByColumns, SearchDirection:=xlPrevious)

    If c Is Nothing Then LastUsedCol = 1 Else LastUsedCol = c.Column

End Function
```

```
Private Function HeaderColumn(ByVal ws As Worksheet, ByVal headerText As
String) As Long

    Dim i As Long, lastCol As Long
```

```

lastCol = LastUsedCol(ws)
For i = 1 To lastCol
    If LCase$(Trim$(CStr(ws.Cells(1, i).Value))) = LCase$(Trim$(headerText)) Then
        HeaderComponent = i
        Exit Function
    End If
Next i
HeaderColumn = 0
End Function

```

```

Private Sub RequireHeader(ByVal ws As Worksheet, ByVal headerText As String)
    If HeaderComponent(ws, headerText) = 0 Then
        Err.Raise vbObjectError + 200, , "Missing required column: " & headerText
    End If
End Sub

```

COMPLETE DESIGN REFERENCE

Property	Value
Font (all)	Aptos, 10pt
Canvas background	RGB(246, 250, 244)
Header gradient	RGB(6,66,25) → RGB(17,112,31), horizontal
Header icon oval	RGB(12,88,30) fill, RGB(230,245,230) border, "\$", size 22
Title text	"EXPENSES DASHBOARD FY 2025", white, bold, size 44
Date text	"Data as of:" + date, white, bold, size 18, WordWrap=True
Column width (1–30)	7.3
Row height rows 1–3	22
Row height rows 4–29	18

Property	Value
Zoom	70%
Gridlines	Hidden
Headings	Visible (True) — set twice: at top of BuildDashboardSheet and at very end
KPI card background	White, border RGB(208,226,203), weight 0.75, shadow on
KPI icon oval	38×38, DiagonalUp gradient, accent→RGB(8,80,25), no border, size 13
KPI value font	Bold, size 28, RGB(5,74,25), formula-linked
KPI title font	Bold, size 18, RGB(45,65,45)
KPI subtitle font	Size 8, RGB(80,95,80)
Primary chart green	RGB(10,93,30)
Secondary chart green	RGB(39,142,41)
Budget bar 2 colour	RGB(166,210,96)
Chart border	RGB(208,226,203), weight 0.7
Chart title colour	RGB(5,74,25), bold, size 12
Data label font	Size 8, bold, RGB(55,55,55)
Number format (money)	#,##0,"K"
Number format (count)	#,##0
Number format (rate)	0.0%
Slicer style	SlicerStyleLight6
Slicer fill	RGB(247,252,245)
Slicer border	RGB(184,215,176)
Month slicer columns	4

Property	Value
Dept slicer columns	6
Hidden columns	AF:AG
Page setup	FitToPagesWide=1, FitToPagesTall=1, Zoom=False
ScreenUpdating	True throughout (live animation visible)
Completion message	MsgBox "Dashboard Created Successfully", vbInformation, "Done"

KEY DIFFERENCES FROM EARLIER VERSION

This v2 module adds/changes the following — all must be reproduced:

1. **ShowProgress + Wait helpers** — status bar animation at every sub-step
 2. **Application.ScreenUpdating = True** — live screen updates visible to user
 3. **DoEvents calls** — interspersed throughout for UI responsiveness
 4. **Zoom = 70** (not 80) — set twice in BuildDashboardSheet
 5. **ActiveWindow.DisplayHeadings = True** — row/column headings permanently ON; set early AND at the very end; NOT set in ApplyDashboardCanvas or FinalDashboardFormatting
 6. **Header shape positions** — calculated dynamically using named Double variables (not inline range references)
 7. **Title textbox** — .TextFrame2.WordWrap = msoFalse, .TextFrame2.AutoSize = msoAutoSizeNone
 8. **Date textbox** — .TextFrame2.WordWrap = msoTrue, font size = 18 (not 16), width = 190, left = bannerLeft+bannerWidth-190-10
 9. **Payment chart legend** — HasLegend = False
 10. **Doughnut legend** — fixed pixel values: Left=229.614, Width=206.285
 11. **Payment chart height** — post-format shrink: co.Height * 0.8986011428
 12. **lo2 variable** — separate Dim lo2 As ListObject for the "already exists" check in PrepareDataSheet
 13. **Completion MsgBox** — shows in CleanExit block on success
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HOW TO USE

1. Open your Excel workbook and ensure the **Data** sheet exists with the required headers.
2. Press Alt+F11 to open the VBA editor.
3. Insert a new standard module (Insert > Module).
4. Paste the complete code into the module.
5. Close the VBA editor.
6. Press Alt+F8, select **BuildExpensesDashboard**, click **Run**.
7. Watch the status bar for live progress. A success message confirms completion.

return the complete VBA code directly in chat. Do NOT provide bas file

